# Course 3 Cluster Analysis

## 3.1. Core Concepts

- **Supervised vs. Unsupervised Learning:**
  - **Supervised Learning (Classification)**: Infers a function from labeled training data consisting of input objects and desired output values. It aims to generalize unseen situations.
  - **Unsupervised Learning (Clustering)**: Infers a structure from "unlabeled" data where no correct output is provided. There is no direct evaluation of accuracy against a ground truth.

- **Cluster Analysis Definition: The task of grouping objects so that objects in the same group are more similar to each other than to those in other groups**. It is a main task of exploratory data mining.

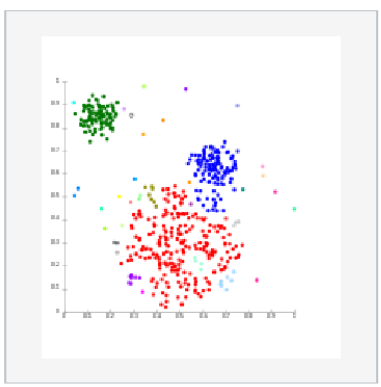
- **Cluster Membership Models:**
  - **Hard (Crisp) Clustering:** Each object belongs to a cluster or not (binary).
  - **Soft (Fuzzy) Clustering:** Each object belongs to each cluster to a certain degree (e.g., likelihood).
  - **Strict Partitioning:** Each object is in exactly one cluster.
  - **Overlapping Clustering:** Objects may belong to more than one cluster.

## 3.2. Clustering Methods Analysis

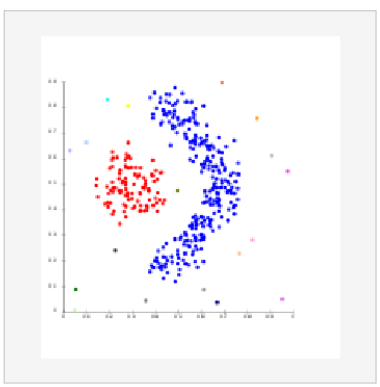
Clustering major models (**definitions**, **advantages**, **disadvantages** and **utility)**:

### A. Connectivity-Based (Connectivity/Hierarchical Clustering)

- **Definition:** Algorithms that connect objects to form clusters based on distance. It produces a hierarchy of clusters (dendrogram) rather than a single partition.
- **Techniques:**
  - **Agglomerative:** Starts with single elements and aggregates them.
  - **Divisive:** Starts with the complete set and divides it.
  - **Linkage Criteria:** ***Single-linkage*** (**minimum distance**), ***Complete-linkage*** (maximum distance), ***UPGMA*** (average linkage).
- **Advantages:**
  - Provides an extensive hierarchy of clusters rather than a single snapshot.
  - ***Considered a theoretical foundation for cluster analysis***.
- **Disadvantages:**
  - **Complexity:** Generally O(n^3) for agglomerative, too slow for large datasets.
  - **Outliers:** Not robust; outliers can cause "chaining phenomena" (merging unrelated clusters).
  - **Ambiguity:** Does not produce a unique partitioning; the user must choose the clusters



Single-linkage on Gaussian data. At 35 clusters, the biggest cluster starts fragmenting into smaller parts, while before it was still connected to the second largest due to the single-link effect.



Single-linkage on density-based clusters. 20 clusters extracted, most of which contain single elements, since linkage clustering does not have a notion of "noise".

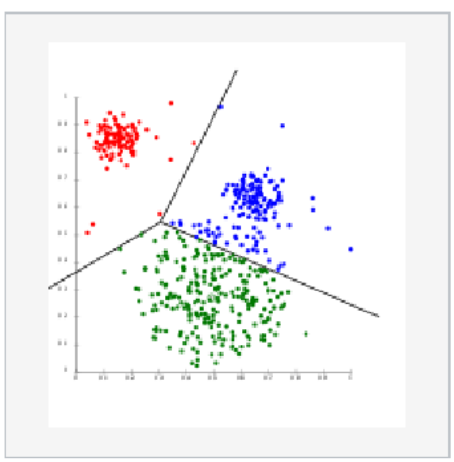
## B. Centroid-Based (k-Means)

- **Definition:** *Clusters are represented by a central vector* (***centroid***). ***The algorithm minimizes the squared distance between objects and their assigned cluster center***.
- **Techniques:** ***k-Means*** (Lloyd's algorithm), ***k-Medoids*** (centroid is a data member), ***k-Means++*** (better initialization), ***Fuzzy C-Means***.
- **Advantages:**
    - Efficient and conceptually close to **nearest-neighbor classification**.

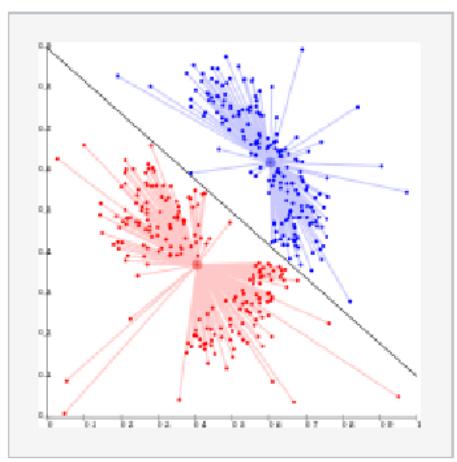
**Note**: *K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for both classification and regression tasks. It is known for its simplicity and effectiveness, relying on the principle that similar data points exist in close proximity to each other*.

    - Partitions space into Voronoi diagrams.
- **Disadvantages:**
    - **Fixed k**: ***The number of clusters*** (***k***) ***must be specified in advance***.
    - **Local Optima:** ***Sensitive to initialization***; often requires multiple runs.
    - **Convexity:** Cannot find non-convex clusters; prefers spherical clusters of similar size.
    - **Boundaries:** Often produces incorrectly cut cluster borders because it optimizes centers, not borders.

***Voronoi Diagrams***:



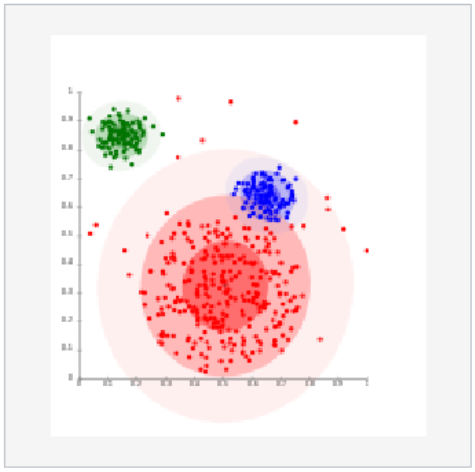
K-means separates data into Voronoi-cells, which assumes equal-sized clusters (not adequate here)



K-means cannot represent density-based clusters

## C. Distribution-Based (Gaussian Mixture Models - GMM)

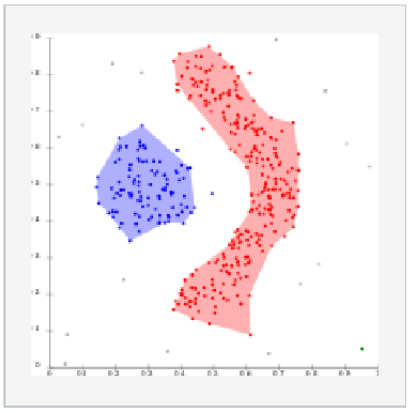
- **Definition:** ***Clusters are defined as objects belonging most likely to the same statistical distribution***. Often uses the Expectation-Maximization (EM) algorithm.
- **Advantages:**
  - **Correlation:** Can capture correlation and dependence between attributes.
  - **Theoretical Foundation:** Excellent mathematical foundation resembling how artificial data is generated.
  - **Soft Clustering:** Naturally supports soft assignments (probabilities).
- **Disadvantages:**
  - **Overfitting:** Suffer from overfitting unless model complexity is constrained.
  - **Complexity:** Difficult to choose appropriate model complexity for real-world data that lacks concise mathematical models.



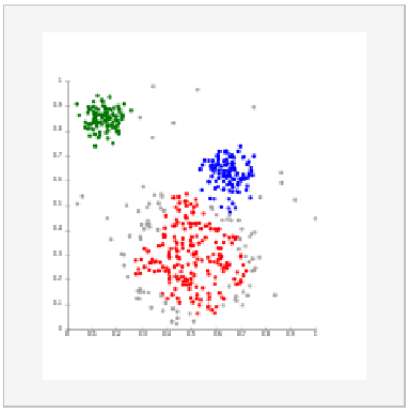
On Gaussian-distributed data, EM works well, since it uses Gaussians for modelling clusters

## D. Density-Based (DBSCAN, OPTICS)

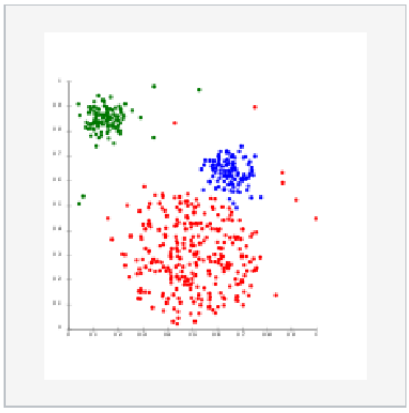
- **Definition:** Clusters are defined as areas of higher density separated by sparse areas.
- **Techniques:**
  - ***DBSCAN*: *Connects points satisfying a density criterion*** (radius ε, MinPts).
  - **OPTICS:** Generalization that removes the need for a fixed range parameter and produces hierarchical results.
  - **Mean-Shift:** Moves objects to local density maxima.
- **Advantages:**
  - **Arbitrary Shapes:** Can discover clusters of arbitrary shapes (unlike k-means).
  - **Noise Handling:** Explicitly handles noise and outliers.
  - **Efficiency:** DBSCAN is efficient (O(n log n)) and deterministic for core points.
- **Disadvantages:**
  - **Varying Density:** Fails on datasets with overlapping distributions or varying densities (e.g., Gaussian mixtures).
  - **Borders:** Cluster borders can look arbitrary as density decreases continuously.



Density-based clustering with DBSCAN.



DBSCAN assumes clusters of similar density, and may have problems separating nearby clusters



OPTICS is a DBSCAN variant that handles different densities much better

## 3. Metrics and Validation Techniques

Validation primarily in the context of classification (Supervised) vs. clustering (Unsupervised), noting that clustering lacks objective accuracy measures against known classes.

### Algorithm Validity

- ***Soundness*:** An algorithm is sound if every returned result is a valid solution.
- ***Completeness*:** An algorithm is complete if every solution can be found by it.

### Classification Metrics (for Supervised Context)

- ***Accuracy*:** The closeness of a measured value to a known standard. Calculated as:

  **{ tp+tn } / { tp+tn+fp+fn }**
- ***Precision*:** The closeness of measurements to each other (independent of accuracy). In prediction terms: "What percent of positive predictions were correct?". Calculated as:

  **{ tp } / { tp+fp }**
- ***Recall*:** "What percent of positive cases did you catch?". Calculated as:

  **{ tp } / { tp+fn }**
- ***True Negative Rate*:** Calculated as

  **{ tn } / { tn+fp }**

### Clustering Issues to Consider

- **Unequal Variance:** Standard algorithms may struggle if clusters have vastly different variances.
- **Irregular Shapes:** Standard centroid models fail on non-spherical, contour, or "shell" shapes.

# 5 Fuzzy Multicriterial Analysis (FMA)

## 5.1. Core Concepts & Motivation

The central theme is Fuzzy Multicriterial Analysis, which introduces ***fuzziness to model data item weights***, ***addressing limitations in traditional*** "***crisp***" ***methods***.

- **The Problem with Traditional Methods**:
  - ***Least-Squares Regression*****:** Assumes constant, independent errors and is highly sensitive to outliers.
  - ***Practical Limitation*****:** Real-world data often includes outliers and non-constant variance, reducing the effectiveness of standard robust techniques.
- **The Fuzzy Solution**:
  - ***Use fuzzy membership degrees to weigh data items***.
  - This allows the model to handle data imperfections and "isolated" points more effectively than binary inclusion/exclusion.

## 5.2 Methods

### Fuzzy Multicriterial Analysis → Fuzzy Regression

- Definition & Algorithm:
  - Concept: It views the regression problem as determining a fuzzy partition {A, $\underline{A}$}, where A is the real data class and $\underline{A}$ is a virtual class with a hypothetical prototype.
  - Objective: Find the fuzzy set A and prototype L (line/curve) that minimizes the objective function J(A,L).
  - Algorithm Steps:
    1. Initialize memberships A(x) = 1.
    2. Compute prototype L minimizing the objective function.
    3. Update memberships A based on distance to L.
    4. Iterate until convergence.

- ***Fuzzy Linear Regression*** (***FLR***):
    - Dissimilarity Measure: Uses the square distance to the line (orthogonal distance), not the vertical distance used in standard regression.
    - Prototype: Determined by the fuzzy mean and the principal component (eigenvector).
    - Polarization: A constant "alpha" sets the polarization of the partition; alpha approx 0.10 typically yields the best results.
- Analysis:
    - **Advantages**: ***Operational in all testing conditions***, ***robust against*** outliers, ***and scale independent*** (***with improvements***).
    - **Utility**: Useful for modeling relationships where data is imprecise or incomplete and detecting data types through repeated runs.

### Fuzzy Multicriterial Analysis → Fuzzy Principal Component Analysis

***Fuzzy PCA  is used for unsupervised dimensionality reduction (ignores labels or input data does not have labels)***. ***The fuzzy variant addresses the issue of points that are isolated with respect to the principal directions***.

- Approaches:
    - ***Fuzzy PCA*** (***First Component***):
        - Problem: Addresses points isolated only regarding the first component.
        - Method: Calculates membership degrees based on the distance to the first component. It computes a "Fuzzy Covariance Matrix" weighted by these memberships.
        - Advantage: Reduces the significance of isolated points (outliers) when calculating the primary axis.
    - **Fuzzy PCA** (**Orthogonal**):
        - Problem: Need to fuzzify all components, not just the first.
        - Method: Recursively projects data into smaller-sized spaces orthogonal to the current fuzzy eigenvector.
        - Advantage: Reduces computation of subsequent components to solving the first component of a smaller matrix.

- Selection Criteria (Dimensionality Reduction):
    - Eigenvalue Value: Keep factors with eigenvalues > 1.
    - Difference: Keep factors until the difference between consecutive eigenvalues drops below a threshold.
    - Cumulative Sum: Keep enough factors to cover a ratio (e.g., 90%) of the total variance.

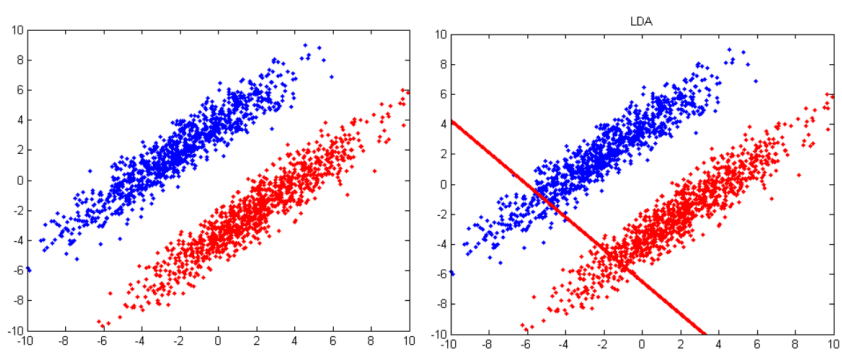
### Fuzzy Multicriterial Analysis → Multi-Dimensional Scaling (MDS)

***MDS maps high-dimensional data to a lower-dimensional metric space while preserving distances***.

- ***Metric MDS*** (***Shepard***): Minimizes the squared error between original distances and projected distances.
- ***Non-metric MDS*** (***Kruskal***): Uses a monotonically increasing function to preserve the rank order of distances rather than exact values.
- ***Sammon Mapping***: A nonlinear mapping that emphasizes preserving local distances.

### Linear Discriminant Analysis (LDA)

***LDA is a supervised statistical technique used for dimensionality reduction and classification***. Unlike PCA, which looks for the axes of maximum variance (unsupervised), **LDA looks for the axes that maximize the separation between multiple classes**.




It projects data onto a lower-dimensional space with the specific goal of maximizing the distance between the means of different classes while minimizing the scatter (variance) within each class.

# 6 Fuzzy Logic Control (FLC)

## 6.1 Core Concepts & Summary

**Definition**:

***A system that manages or directs other devices to achieve a desired result*** (***like a thermostat keeping a room at a set temperature***).

**Closed-Loop Control (Feedback)**:

The system looks at the actual output (e.g., the current room temp) and compares it to the target. It uses "feedback" to correct errors.

**Fuzzy Control**:

Instead of using complex calculus (differential equations) like traditional controllers, Fuzzy Control uses "If-Then" rules based on human experience.

- Simplified: Traditional control is like a calculator doing math to drive a car. Fuzzy control is like a human driver saying, "If the curve is sharp, turn the wheel hard."

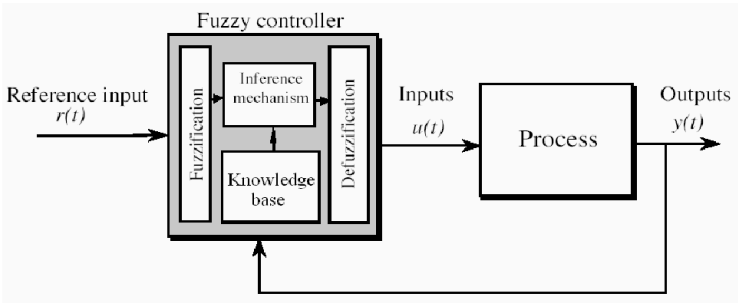
## 6.2 The 4 Main Steps of Fuzzy Control (Architecture)

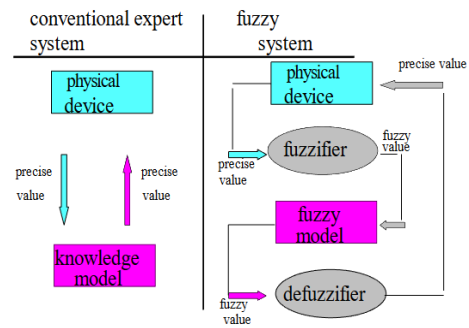
The "Fuzzy Controller" sits between the sensors (inputs) and the machine (outputs).

It works in four stages:

- ***Step 1***: ***Fuzzification*** (***The Translator***)
  - Concept: Takes the "crisp" numbers from sensors (e.g., "Speed is 55 mph") and converts them into fuzzy degrees (e.g., "Speed is 0.6 Fast").
  - Simplified: ***It translates machine numbers into human concepts***.
- ***Step 2***: ***Knowledge Base*** (***The Rule Book***)
  - Concept: *Stores the* "*If-Then*" *rules provided by experts*.
  - Example: "IF temperature is Hot AND pressure is Rising, THEN fan speed is High."
- **Step 3**: **Inference Engine** (**The Brain**)
  - Concept: It looks at the fuzzy inputs and the rule book to decide what to do. It figures out which rules apply and combines them to get a "fuzzy" result.
  - Simplified: ***It weighs the evidence***. If the rule says "turn slightly left" and another says "turn hard left," it combines these opinions.

- ***Step 4***: ***Defuzzification*** (***The Converter***)
    - Concept: Takes the combined fuzzy result (a vague shape or distribution) and calculates a single "crisp" number to send to the machine.
    - Simplified: The machine can't understand "somewhat fast." This step picks a specific number (like "60 mph") that best represents the fuzzy decision.
    - Methods: The most common method is Center of Gravity (CoG), which finds the balance point of the fuzzy result.



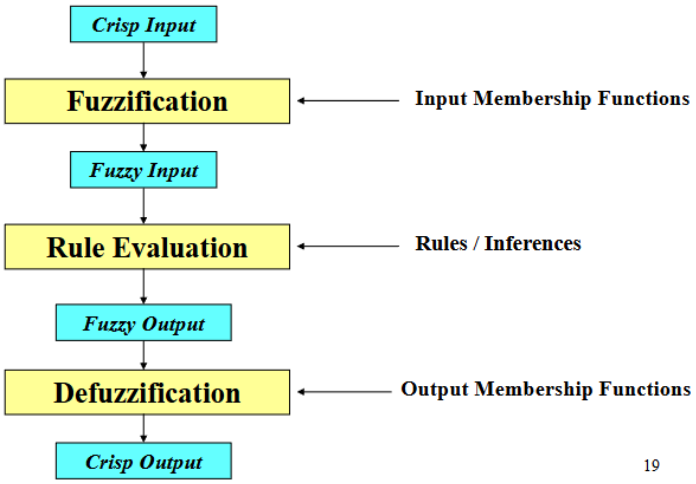

In order to process the input to get the output reasoning there are six steps involved in the ***creation of a rule based fuzzy system***:

1. Identify the inputs and their ranges and name them;
2. Identify the outputs and their ranges and name them;
3. Create the degree of fuzzy membership function for each input and output;
4. Construct the rule base that the system will operate under;
5. Decide how the action will be executed by assigning strengths to the rules;
6. Combine the rules and defuzzify the output.

## 6.3 Fuzzy Models

### Mamdani Model (Human-Like Model)

- **Concept**: It is designed to capture expert knowledge directly. Because both the "If" (input) and "Then" (output) parts of the rule are defined by fuzzy sets (words), ***it mimics human reasoning***.
- **Rule Structure**:
    1. **Format**: *IF <input> is <Linguistic Term> THEN <output> is <Linguistic Term>*.
    2. Example: "IF Pressure is High, THEN Valve is OpenFast."
- **How it Works** (**Inference**):
    1. **Fuzzify Inputs**: Convert sensor readings into degrees of membership (e.g., Pressure is 0.7 High).
    2. **Apply Implications**: The system typically uses the "Minimum" operator to clip the output fuzzy set. If the rule only fires at 0.7 strength, the output shape "Open_Fast" is chopped off at the 0.7 level.
    3. **Aggregation**: All the clipped fuzzy shapes from all active rules are combined (using the "Max" operator) into one single, complex fuzzy shape.
    4. **Defuzzification**: This complex shape must be converted into a crisp number using methods like the Centroid (Center of Gravity) method.
- **Advantages**: Widespread acceptance and highly interpretable by humans.
- **Disadvantages**: Computationally expensive because calculating the area/centroid of complex, irregular fuzzy shapes takes processing power.



## Takagi-Sugeno (TS) Model ("The Mathematical One")

***This model was developed to create a systematic way to generate fuzzy rules from input-output datasets**, **rather than just relying on human intuition**.*

- **Concept**: It keeps the fuzzy inputs (to handle uncertainty) but replaces the fuzzy outputs with precise mathematical formulas. **This makes it a hybrid between fuzzy logic and traditional control math**.
- **Rule Structure**: IF <input> is <Linguistic Term> THEN <output> is z = f(x).
    - Zero-Order TSK: The output is a constant number (e.g., THEN z = 5).
    - First-Order TSK: The output is a linear equation (e.g., THEN z = 2x + 3).
- **How it Works** (**Inference**):
    - Fuzzify Inputs: Same as Mamdani (calculate rule strength, e.g., $w_1 = 0.7$).
    - Calculate Outputs: Instead of clipping shapes, the system simply calculates the math formula for that rule.
    - Defuzzification (Weighted Average): The final output is calculated immediately by averaging the results of all rules, weighted by their strength.
- **Advantages**:
    - ***Computationally Efficient***: No complex integration or area calculations needed; it's just simple algebra.
    - ***Optimization***: Works very well with adaptive techniques (like neural networks) and optimization algorithms.
    - ***Continuity***: Guarantees a smooth, continuous output surface.
- **Disadvantages**: Less interpretable. A rule saying THEN z = 0.5x + 4.2 is harder for a human operator to intuitively grasp than THEN Valve is Medium.

| **Feature** | **Mamdani** | **Takagi-Sugeno (TS)** |
|---|---|---|
| Output Type | Fuzzy Set (Linguistic terms like "Hot") | Mathematical Function (Linear or Constant) |
| Defuzzification | Required (e.g., Centroid calculation) | Built-in (Weighted Average) |
| Interpretability | High (Human-readable rules) | Low (Mathematical rules) |
| Speed | Slower (Complex math for area) | Faster (Simple algebra) |
| Best Use Case | Capturing expert human knowledge | Control systems requiring efficiency & data fitting |

## 6.4 Fuzzy Sets

**Core Concept: Fuzzy Sets are the foundational building blocks of Fuzzy Logic**. Unlike classical (crisp) sets where an element either belongs (1) or does not belong (0), fuzzy sets allow for degrees of membership [0 - 1].

- ***Partial Truth***: Fuzzy logic deals with "degrees of truth," allowing a statement to be partially true and partially false at the same time.
- ***Membership***: Everything is a matter of degree. A value can have a membership degree anywhere between 0 and 1.
- ***Linguistic Variables***: Fuzzy sets are often described using natural language (linguistic variables) rather than precise numbers. For example, in a system analyzing service quality, the variable "Service" might have fuzzy sets like "Poor," "Good," and "Excellent".
- ***The Membership Function***: This is the curve that defines how each point in the input space maps to a membership value (or degree of truth) between 0 and 1.

**Advantages:**

- **Conceptual Simplicity**: The mathematical concepts behind fuzzy reasoning are very simple and easy to understand.
- **Flexibility**: It is easy to modify a system by simply adding more rules or logic layers.
- **Tolerance for Imprecision**: Fuzzy logic is tolerant of imprecise or noisy data.
- **Modeling Nonlinearity**: It can model arbitrary nonlinear functions of arbitrary complexity, which is difficult for standard linear controllers.
- **Leverages Expert Experience**: It can be built directly on the experience of experts without needing a precise mathematical training model.
- **Natural Language**: Because it is based on natural language, the logic is easy for humans to understand and communicate.

**Disadvantages & Limitations:**

- Not a Universal Solution: Fuzzy logic is not a cure-all and should not be used for every problem.
- Inefficiency for Simple Problems: If a simpler solution (like a standard mathematical model or PID controller) already exists or is easy to derive, fuzzy logic is unnecessary and might be "overkill".
- Approximation: It is essentially a convenient way to map input spaces to output spaces, implying it may lack the rigorous exactness required for some purely analytical problems.

**Use Cases**: ***Fuzzy Sets are used extensively in control systems where the process is complex or defined by human intuition rather than equations***.

- Complex Control Systems: Ideal for systems that are difficult to model mathematically, are nonlinear, or contain significant uncertainty.
- Consumer Electronics: Widely used in cameras, camcorders, washing machines, and microwave ovens.
- Industrial Automation: Applied in precision mechanics and industrial manufacturing processes.
- Automotive: Used in vehicle subsystems like engine control, automatic transmissions, and ABS systems.
- Decision Making (Example): A "Tipping" system that calculates the tip percentage based on fuzzy inputs of "Service Quality" (Poor/Good/Excellent) and "Food Quality" (Rancid/Delicious).
- Environment Control: Controlling HVAC (Heating, Ventilation, and Air Conditioning) systems, such as adjusting a shower temperature.

# 7 Rough Sets

## 7.1 Rough Sets

**Core Concept**: ***Rough Set theory offers a mathematical approach to handling vagueness, uncertainty and pattern discovery in data***. Unlike standard set theory, where an object either belongs to a set or it does not, Rough Sets handle situations where concepts cannot be sharply defined.

- ***Indiscernibility***: ***This is the foundational concept of Rough Sets***. It arises from the idea that if two objects have the exact same values for a specific set of attributes (like symptoms or physical measurements), they are indistinguishable from one another with respect to those attributes. These indistinguishable objects form "equivalence classes" or elementary building blocks of knowledge.
- **Set Approximations**: **Because some concepts cannot be defined crisply due to limited information** (**indiscernibility**), **Rough Sets approximate a concept using two precise boundaries**:
    - **Lower Approximation**: This consists of all objects that can be classified with certainty as belonging to the concept based on the available attributes.
    - **Upper Approximation**: This includes all objects that can possibly belong to the concept; these are objects that have a non-empty intersection with the target concept.
        - **Boundary Region**: This is the difference between the upper and lower approximations. Objects in this region cannot be decisively classified as members or non-members of the set using the available data. A set is considered "rough" if this boundary region is not empty.
- **Reducts and Core**:
    - **Reducts**: In many datasets, some attributes are redundant. A "reduct" is a minimal subset of attributes that preserves the original ability to classify objects (maintains the indiscernibility relation). It allows for data reduction without losing essential information.
    - **Core**: The "core" is the set of attributes that are absolutely indispensable. These attributes appear in every possible reduct and cannot be removed without deteriorating the classification quality.

- **Rough Membership**: Unlike fuzzy sets where membership is a primary, often subjective concept, rough membership is calculated based on data. It quantifies the degree of overlap between an equivalence class and the target concept, essentially acting as a frequency-based estimate of probability.

## 7.2 The Rough Set Knowledge Discovery in Databases Process

### *7.2.1 Rough Sets KDD*

Rough Sets constitute a sound basis for Knowledge Discovery in Databases (KDD). The process generally involves several key stages designed to handle real-world issues like large datasets and mixed data types.

1. ***Discretization***: Real-world data often contains continuous values (e.g., temperature). Rough set methods convert these continuous values into discrete intervals (cuts) to make patterns visible and manageable.
2. ***Attribute Selection***: This step aims to find an optimal subset of attributes. By using heuristics combined with the concepts of Reducts and Core, the process removes redundant data to improve the efficiency and accuracy of rule discovery.
3. ***Rule Discovery***: The final stage involves extracting patterns or rules (e.g., "If Symptom X and Symptom Y, then Diagnosis Z"). Systems like GDT-RS (Generalization Distribution Table) use rough set principles to discover rules that can predict outcomes for new, unseen instances.

### *7.2.2 Advantages*:

- ***Objective Handling of Uncertainty***: Rough sets provide a mathematical tool to handle vagueness and uncertainty without relying on subjective membership functions, which is a key difference from Fuzzy Set theory.
- ***Data Reduction***: The theory provides powerful methods (Reducts) for feature selection and data reduction. It can identify and remove superfluous attributes, which simplifies the data without losing critical classification information.
- ***Pattern Discovery in Noisy Data***: The methodology is based on the premise that lowering the degree of precision can make data patterns more visible, making it effective for finding patterns hidden in noisy or incomplete data.
- ***Explicit Rule Uncertainty***: When discovering rules, the uncertainty and stability of a rule (its ability to predict unseen data) can be explicitly represented and quantified.

### *7.2.3 Disadvantages*:

- ***Need for Discretization***: Rough set theory primarily deals with symbolic or discrete data. Continuous valued attributes (real numbers) must undergo discretization (dividing values into intervals) before analysis, which is a complex process where the choice of method depends heavily on the data.
- ***Computational Complexity***:
    - Finding the optimal reduct (minimal set of attributes) is computationally expensive. The number of possible reducts can be exponential relative to the number of attributes, making exhaustive search impractical for large datasets.
    - To mitigate this, heuristic algorithms are often required to find sub-optimal solutions efficiently, as exact solutions have high time complexity.

### *7.2.4 Use Cases*

- Medical Diagnosis: Bacterial Examination: Used to analyze relationships between detected bacteria and antibiotic sensitivity;
- Geological Safety: Slope Collapse: Analysis of geological data (altitude, soil thickness, fault lines) to determine the causes of slope collapses and predict dangerous areas;
- **General Data Mining**: **Broadly applicable for feature extraction, template generation and association rule generation in large databases.**

## 7.3 Decision Trees

### 7.3.1 Core Concept

***Decision Trees are a supervised learning technique used primarily for classification tasks***. They utilize a "divide-and-conquer" approach to model data.

- **Tree Structure**: The model is represented as a tree where each internal node tests a specific attribute of the data. Each branch extending from a node corresponds to a possible value of that attribute and each leaf node assigns a final class label or prediction.
- **Interpretation**: A DT can be viewed as a tree-structured plan of questions. To classify a new instance, one starts at the root and follows the branches corresponding to the instance's attribute values until a leaf is reached.

### 7.3.2 Decision Boundaries

- **Rectilinear Boundaries**: In single-attribute-test trees, the decision boundaries created in the feature space are rectilinear, meaning they are always parallel to the coordinate axes. This contrasts with other methods (like 1-Nearest Neighbor) which create more complex, Voronoi-style partitions.
- **Oblique Trees**: While standard trees use single attributes, "oblique" trees can test multiple attributes simultaneously (e.g., $x+y < 1$), though finding the optimal split for these is computationally expensive.

### 7.3.3. Construction Algorithm (ID3) The tree is built top-down recursively.

- **Selection**: An attribute is selected for the root node, and a branch is created for each possible value.
- **Splitting**: Training instances are split into subsets based on these branches.
- **Recursion**: The process repeats for each branch using only the instances that reach it.
- **Stopping Criteria**: The process stops when all instances in a node belong to the same class (creating a leaf) , or if no attributes remain to distinguish the instances (creating a leaf with the majority class).

### 7.3.4 Handling Complex Data

- **Numeric Attributes**: Standard algorithms (like ID3) are designed for nominal (categorical) data. For numeric attributes, values are sorted, and adjacent examples with different classes are identified to generate "split points" (e.g., temperature < 70). This effectively converts a continuous range into a binary split.
- **Missing Values**: Strategies for handling missing data include ignoring those instances, assuming the most common value, or, more sophisticatedly, assigning probabilities to possible values and distributing the instance across branches fractionally (used in C4.5).

### 7.3.5 Overfitting and Pruning

Decision trees often grow deep enough to perfectly classify training data, which can lead to overfitting, where the model fits noise or coincidental regularities in the training set and fails to generalize to new data.

Strategies to Prevent Overfitting:

- ***Pre-pruning***: **Stop growing the DT early before it perfectly classifies training data**;
- ***Post-pruning***: **Allow the tree to grow fully, then prune it back**. This is generally more successful in practice.

| ***Advantages*** | ***Disadvantages*** |
|---|---|
| ***Easy to Interpret***: The resulting model is easy for humans to read and understand, as it mimics a series of questions. | ***Sub-tree Replication***: The same sub-tree structure may need to be repeated at different branches, making the tree unnecessarily large. |
| ***Efficiency***: They are computationally efficient to build and classify new examples. | ***Rectilinear Boundaries***: They struggle to model diagonal or complex decision boundaries efficiently because they only test one attribute at a time. |
| ***Robust to Noise***: Through pruning, they can handle noisy data effectively. | ***Overfitting***: Without pruning, they tend to overfit, especially with small training sets or noise. |
| ***Robust to Redundancy***: They handle redundant attributes well (only one will be selected). | ***Irrelevant Attributes***: They are less robust to irrelevant attributes, which might be selected and bloat the tree. |